TTG continuation: partial performance study

LittleX and IssueDesk | all six policies | n=3 independent workflows per cell

48 fresh workflow trials

12 excluded warmups + 36 measured workflows

Primary metric: actual SQL send -> ReadyForQuery completion wall time.

Summed DB calls, interval union, SQL data/control counts and real PID overlap.

Identical seeded workflows and independent response/state correctness guards.

Scope is two applications; the full application and paper tiers remain separate.

LittleX: DB time and SQL counts

Median [IQR]; three fresh measured workflows for every row

Policy	DB sum ms [IQR]	DB union ms [IQR]	SQL total/data/control/other	FG/BG distinct backend overlap ms
Off	39.859 [1.236]	39.859 [1.236]	153/90/63/0	0.000
Legacy TTG / thread	116.583 [1.556]	116.400 [1.532]	144/74/70/0	0.141
Always / thread	50.362 [8.214]	43.831 [7.562]	167/90/77/0	0.000
Always / process	49.514 [2.891]	49.514 [2.891]	159/90/69/0	0.000
Expected benefit / thread	47.846 [2.153]	42.873 [1.440]	165/90/75/0	0.000
Expected benefit / process	50.053 [1.206]	50.053 [1.206]	159/90/69/0	0.000

Overlap column = foreground/background protocol intervals on distinct PostgreSQL backends.

SQL counts are dispatches; actual completion cycles and all raw samples are in the data export.

IssueDesk: DB time and SQL counts

Median [IQR]; three fresh measured workflows for every row

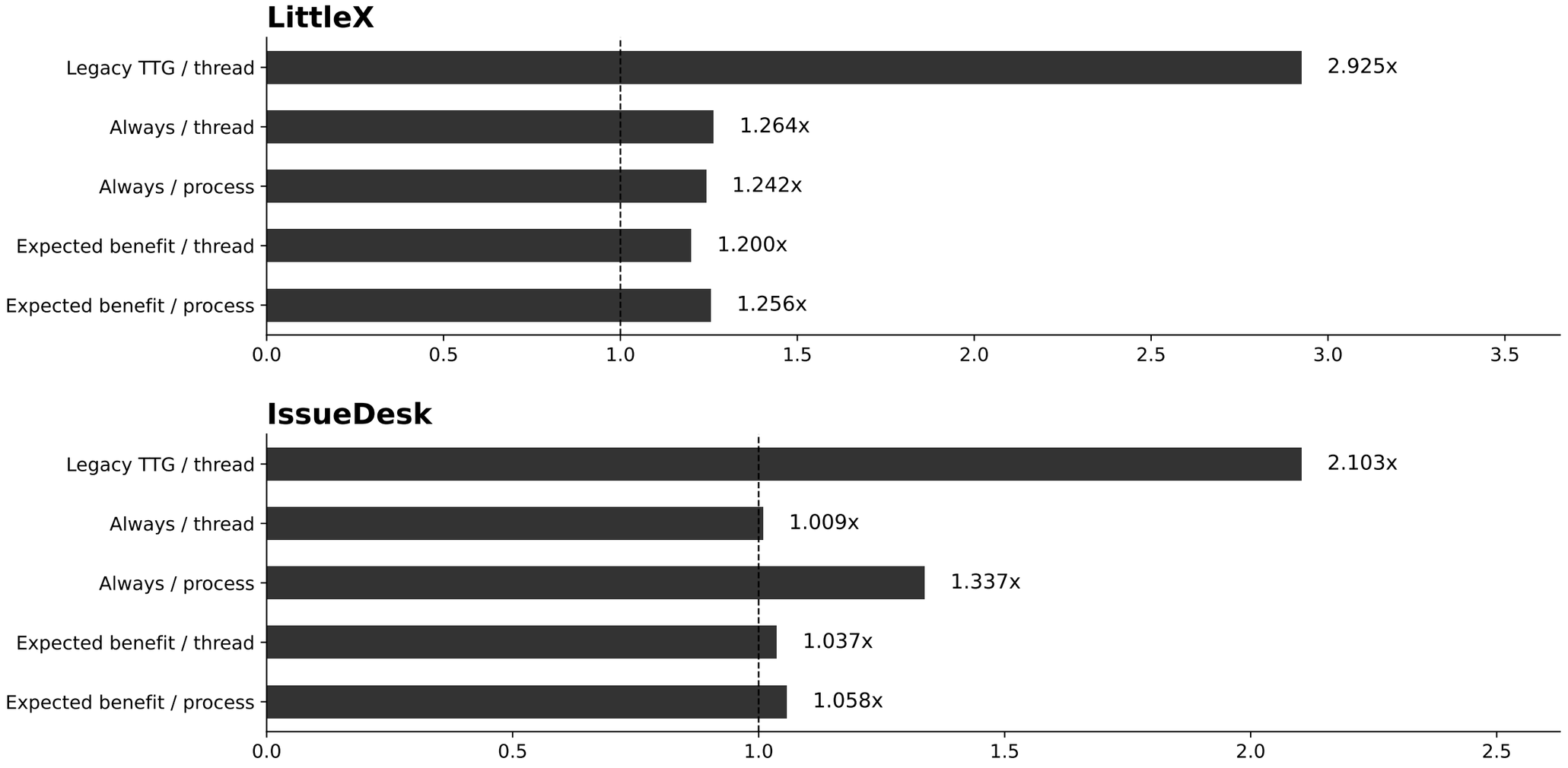
Policy	DB sum ms [IQR]	DB union ms [IQR]	SQL total/data/control/other	FG/BG distinct backend overlap ms
Off	140.773 [11.417]	140.773 [11.417]	555/366/189/0	0.000
Legacy TTG / thread	296.063 [16.585]	293.819 [16.541]	569/354/215/0	3.043
Always / thread	142.029 [2.018]	142.029 [2.018]	563/366/197/0	0.000
Always / process	188.234 [24.292]	188.234 [24.292]	559/366/193/0	0.000
Expected benefit / thread	145.973 [5.113]	145.973 [5.113]	561/366/195/0	0.000
Expected benefit / process	148.872 [0.697]	148.872 [0.697]	558/366/192/0	0.000

Overlap column = foreground/background protocol intervals on distinct PostgreSQL backends.

SQL counts are dispatches; actual completion cycles and all raw samples are in the data export.

Relative summed DB-call time

Ratio of workflow medians to off; <1 lower, >1 higher. Dispersion is shown in the timing tables.



Measurement and provenance

Actual DB evidence; independent correctness and exact source binding

- Send -> ReadyForQuery completion includes the response and client decoding/progress.
- Sum/union gaps cover all overlapping calls; foreground/background overlap is separate.
- Each fresh seeded workflow starts a fresh profile and learns naturally within the trial.
- Setup, authentication, warmup and final persistence verification have separate phases.
- Late SQL, process readiness/startup and IPC phases retain actual request/scope identities.
- Measured SQL logging is off; counts/row bytes and independent controls are hash-bound.

Accepted export SHA256: 17922e61bcf4ff4ca6814a1b23cd758d0ac371d979f1537ab95dc802233c2f4b

Main authorization SHA256: d1fc7443300f79ce5bc2185f900182466d60bed336906d99c54540c585d1dc02

LittleX: measured SQL contributions

Actual primary traces | payload SELECT fingerprint b0075df9e735 | counts per workflow

Policy	Payload SELECTs: FG / BG	Returned rows: FG / BG	BG DISCARD ALL
Off	52 / 0	108 / 0	0
Legacy TTG / thread	35 / 7	35 / 73	7
Always / thread	38 / 14	45 / 63	14
Always / process	38 / 14	45 / 63	6
Expected benefit / thread	40 / 12	58 / 50	12
Expected benefit / process	39 / 13	46 / 62	6

Payload reads move between roles; off and all four continuation modes keep 52 SELECTs / 108 returned rows.

Legacy foreground WITH RECURSIVE fingerprint 9a54ec61abbb: 4 calls in each measured workflow.

Its protocol contributions by repeat: 60.043 / 59.736 / 60.539 ms.

Pooled across three workflows: 12 calls, 180.318 ms.

LittleX: observed overlap and continuation activity

Measured primary traces | exact interval sweep and native scope events

Sum minus union (ms)	Repeat0	Repeat1	Repeat2
Always / thread	6.531	6.289	7.592
Expected benefit / thread	4.139	5.565	4.973

Each gap is fully accounted for by background/background intervals on distinct backends.

Example: PostgreSQL backends 1456352 / 1456356, client 1456185.

Recorded foreground/background overlap in all four continuation configurations: 0 ms.

Always/thread repeat0: 9 scopes, 14 plans including 4 multi-hop plans.

86 native continuation snapshots; reached barrier at gather:277 (here.to_view()).

Full scope sequence, decisions/probes, payload categories and interval/PID pairs are in attribution.json.